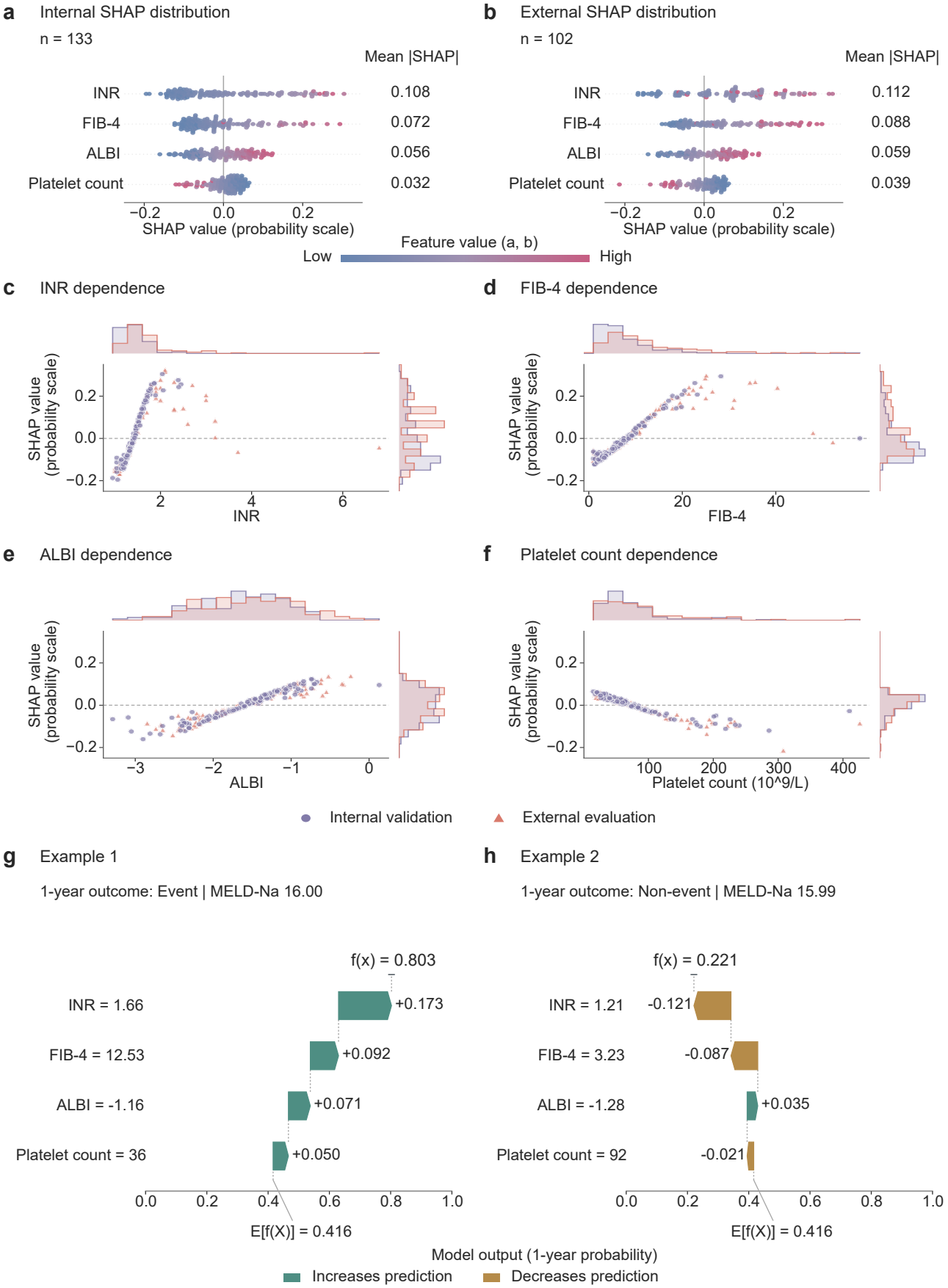

a
Internal SHAP distribution
n = 133
Mean |SHAP|
INR
FIB-4
ALBI
Platelet count
0.108
0.072
0.056
0.032
−0.2
0.0
0.2
SHAP value (probability scale)
b
External SHAP distribution
n = 102
Mean |SHAP|
INR
FIB-4
ALBI
Platelet count
0.112
0.088
0.059
0.039
−0.2
0.0
0.2
SHAP value (probability scale)
Feature value (a, b)
Low
High
c
INR dependence
SHAP value (probability scale)
0.2
0.0
−0.2
2
4
6
INR
d
FIB-4 dependence
SHAP value (probability scale)
0.2
0.0
−0.2
0
20
40
FIB-4
e
ALBI dependence
SHAP value (probability scale)
0.2
0.0
−0.2
−3
−2
−1
0
ALBI
f
Platelet count dependence
SHAP value (probability scale)
0.2
0.0
−0.2
100
200
300
400
Platelet count (10^9/L)
Internal validation
External evaluation
g
Example 1
1-year outcome: Event | MELD-Na 16.00
f(x) = 0.803
INR = 1.66
+0.173
FIB-4 = 12.53
+0.092
ALBI = -1.16
+0.071
Platelet count = 36
+0.050
0.0
0.2
0.4
0.6
0.8
1.0
E[f(X)] = 0.416
h
Example 2
1-year outcome: Non-event | MELD-Na 15.99
f(x) = 0.221
INR = 1.21
-0.121
FIB-4 = 3.23
-0.087
ALBI = -1.28
+0.035
Platelet count = 92
-0.021
0.0
0.2
0.4
0.6
0.8
1.0
E[f(X)] = 0.416
Model output (1-year probability)
Increases prediction
Decreases prediction